

Cheat Sheet: Data Preprocessing Tasks in Pandas & Plot Libraries

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Review this cheat sheet to learn more about the various data preprocessing tasks in Pandas, their descriptions, syntax, and examples.

Cheat Sheet : Data Preprocessing Tasks in Pandas

Load CSV data

Read data from a CSV file into a Pandas DataFrame

Syntax:

```
1 pd.read_csv('filename.csv')
```

Example:

```
1 df_can=pd.read_csv('data.csv')
```

Handling Missing Values

Drop rows with missing values

Syntax:

```
1 df.dropna()
```

Example:

```
1 df_can.dropna()
```

Handling Missing Values

Fill missing values with a specified value

Syntax:

```
1 df.fillna(value)
```

Example:

```
1 df_can.fillna(0)
```

Removing Duplicates

Remove duplicate rows

Syntax:

```
1 df.drop_duplicates()
```

Example:

```
1 df_can.drop_duplicates()
```

Renaming Columns

Rename one or more columns

Syntax:

```
1 df.rename(columns={'old_name': 'new_name'})
```

Example:

```
1 df_can.rename(columns={'Age': 'Years'})
```

Selecting Columns

Select a single column

Syntax:

```
1 df['column_name'] or df.column_name
```

Example:

```
1 df_can.Age or df_can['Age']
```

Selecting Columns

Select multiple columns

Syntax:

```
1 df[['col1', 'col2']]
```

Example:

```
1 df_can[['Name', 'Age']]
```

Filtering Rows

Filter rows based on a condition

Syntax:

```
1 df[df['column'] > value]
```

Example:

```
1 df_can[df_can['Age'] > 30]
```

Applying Functions to Columns

Apply a function to transform values in a column

Syntax:

```
1 df['column'].apply(function_name)
```

Example:

```
1 df_can['Age'].apply(lambda x: x + 1)
```

Creating New Columns

Create a new column with values derived from existing ones

Syntax:

```
1 df['new_column'] = expression
```

Example:

```
1 df_can['Total'] = df_can['Quantity'] * df_can['Price']
```

Grouping and Aggregating

Group rows by a column and apply aggregate functions

Syntax:

```
1 df.groupby('column').agg({'col1': 'sum', 'col2': 'mean'})
```

Example:

```
1 df_can.groupby('Category').agg({'Total': 'mean'})
```

Sorting Rows

Sort rows based on a column

Syntax:

```
1 df.sort_values('column', ascending=True/False)
```

Example:

```
1 df_can.sort_values('Date', ascending=True)
```

Displaying First n Rows

Show the first n rows of the DataFrame

Syntax:

```
1 df.head(n)
```

Example:

```
1 df_can.head(3)
```

Displaying Last n Rows

Show the last n rows of the DataFrame

Syntax:

```
1 df.tail(n)
```

Example:

```
1 df_can.tail(3)
```

Checking for Null Values

Check for null values in the DataFrame

Syntax:

```
1 df.isnull()
```

Example:

```
1 df_can.isnull()
```

Selecting Rows by Index

Select rows based on integer index

Syntax:

```
1 df.iloc[index]
```

Example:

```
1 df_can.iloc[3]
```


Selecting Rows by Index

Select rows in a specified range

Syntax:

1	<code>df.iloc[start:end]</code>

Example:

1	<code>df_can.iloc[2:5]</code>

Selecting Rows by Label

Select rows based on label/index name

Syntax:

1	<code>df.loc[label]</code>

Example:

1	<code>df_can.loc['Label']</code>

Selecting Rows by Label

Select rows in a specified label/index range

Syntax:

1	<code>df.loc[start:end]</code>

Example:

1	<code>df_can.loc['Age': 'Quantity']</code>

Summary Statistics

Generates descriptive statistics for numerical columns

Syntax:

1	<code>df.describe()</code>

Example:

1	<code>df_can.describe()</code>

Review this cheat sheet to learn more about the various plot libraries, their main purpose, key features, programming languages, level of customization, dashboard capabilities, and the types of plots possible in them.

Cheat Sheet : Plot Libraries

Library	Main Purpose	Key Features	Programming Language	Level of Customization	Dashboard Capabilities
Matplotlib	General-purpose plotting	Comprehensive plot types and variety of customization options	Python	High	Requires additional components and customization
Pandas	Fundamentally used for data manipulation but also has plotting functionality	Easy to plot directly on Panda data structures	Python	Medium	Can be combined with web frameworks for creating dashboards
Seaborn	Statistical data visualization	Stylish, specialized statistical plot types	Python	Medium	Can be combined with other libraries to display plots on dashboards
Plotly	Interactive data visualization	interactive web-based visualizations	Python, R, JavaScript	High	Dash framework dedicated for building interactive dashboards
Folium	Geospatial data visualization	Interactive, customizable maps	Python	Medium	For incorporating maps into dashboards, it can be integrated with other frameworks/libraries
PyWaffle	Plotting Waffle charts	Waffle charts	Python	Low	Can be combined with other libraries to display waffle chart on dashboards



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